

Equality in the local clique cover–chromatic bound for graphs of orders two through eight

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Abstract

We give an exact computer-assisted classification of every finite simple graph G of order $2 \leq n \leq 8$ satisfying $\text{lcc}(G) + \chi(G) = n + 1$, where lcc is the minimum maximum vertex incidence in an edge-clique cover. Within this range, equality holds precisely when G has a universal vertex u and $H = G - u$ satisfies $\chi(H) + \chi(\overline{H}) = |V(H)| + 1$. The numbers of equality isomorphism classes are 1, 2, 4, 8, 18, 40, 98. The proof covers all 13 597 graph classes, certifies exact optima on all 1 252 classes with a universal vertex, and gives strict upper witnesses for the other 12 345 classes. We provide all 171 equality representatives, their exact bounds, and replayable certificates. The computation uses every clique, permits overlapping covers, and makes no assertion for orders at least nine or about bibliographic priority.

1 Statement and scope

All graphs here are finite, simple, and undirected. An *edge-clique cover* of G is a family $\mathcal{C}$ of cliques containing every edge of G . Define

$$\text{lcc}(G) = \min_{\mathcal{C}} \max_{v \in V(G)} |\{Q \in \mathcal{C} : v \in Q\}|.$$

The family need not partition the edges: cliques and covered edges may overlap. Repeated cliques and singleton cliques can be removed without increasing the objective, so all subsets inducing cliques of size at least two form the optimization domain. Restriction to maximal cliques is not valid in general, since enlarging a clique can increase the loads of additional vertices. For an edgeless graph we set $\text{lcc}(G) = 0$. The usual vertex chromatic number is $\chi(G)$. A vertex is *universal* if it is adjacent to every other vertex; a graph with such a vertex is called a *cone* and is written $K_1 \vee H$.

Davoodi’s preprint [1, Theorems 1.1–1.2 and Corollary 1.3] states the general inequality $\text{lcc}(G) + \chi(G) \leq |V(G)| + 1$ and proves that an equality graph of order at least two has a universal vertex. Its examples include complete graphs, nontrivial stars, and $K_n - e$; Section 6 discusses the remaining general equality characterization. The contribution here relative to that inspected source is a complete finite classification, together with a numerical criterion on the graph obtained by deleting a universal vertex. The finite proof below does not depend on the general inequality or the universal-vertex theorem in that source.

Theorem 1 (Exact finite classification). *Let G have order $2 \leq n \leq 8$. Then*

$$\begin{aligned} \text{lcc}(G) + \chi(G) = n + 1 \\ \iff G \text{ has a universal vertex } u \text{ and, for } H = G - u, \\ \chi(H) + \chi(\overline{H}) = |V(H)| + 1. \end{aligned} \tag{1}$$

The equality isomorphism-class counts, in order $n = 2, \dots, 8$, are 1, 2, 4, 8, 18, 40, 98, totaling 171. Appendix A lists one representative of every equality class and its exact χ and lcc values. The accompanying certificate data give matching upper and lower bounds for every listed representative.

The displayed condition on H is the chromatic Nordhaus–Gaddum equality condition, used solely as an explicit numerical predicate. No external structural classification of those equality graphs is assumed. If u and v are both universal, their transposition is an automorphism of G , so deleting either gives isomorphic residual graphs. Thus the predicate does not depend on which universal vertex is chosen.

Table 1: Complete finite domain and equality counts. The final column gives counts in increasing order of $\chi = 2, \dots, n$.

n	All classes	Cones	Equality	Counts by χ
2	2	1	1	1
3	4	2	2	1, 1
4	11	4	4	1, 2, 1
5	34	11	8	1, 3, 3, 1
6	156	34	18	1, 4, 8, 4, 1
7	1044	156	40	1, 5, 14, 14, 5, 1
8	12346	1044	98	1, 6, 23, 38, 23, 6, 1
Total	13597	1252	171	

This resolves the finite classification at the stated orders. The all-order equality problem is not resolved here, and no extension of (1) to $n \geq 9$ is claimed.

2 Elementary certificate principles

2.1 Exact coloring and clique partitions

For $S \subseteq V(G)$, let $c(S) = \chi(G[S])$ and let $p(S)$ be the minimum number of cliques partitioning S . Singletons are allowed in these vertex partitions. We have $c(\emptyset) = p(\emptyset) = 0$ and $p(S) = \chi(G[S])$ directly from the definitions.

Lemma 2. *For nonempty S , choose its least vertex v in a fixed ordering. Then*

$$c(S) = 1 + \min_{\substack{I \subseteq S, v \in I \\ I \text{ independent}}} c(S \setminus I), \quad (2)$$

$$p(S) = 1 + \min_{\substack{Q \subseteq S, v \in Q \\ Q \text{ a clique}}} p(S \setminus Q). \quad (3)$$

Consequently, checking these recurrences on all subsets certifies both tables using integer arithmetic.

Proof. Every partition of S into independent sets has exactly one part I containing v . Removing it leaves a partition of $S \setminus I$, proving one inequality in (2). Conversely, adjoining any eligible I to an optimal partition of its complement proves the other. The argument for cliques is identical. All remainders are proper subsets, so induction from the empty set determines the tables uniquely. $\square$

Lemma 3 (Neighborhood lower bound). *Every vertex v satisfies $\text{lcc}(G) \geq p(N(v))$.*

Proof. The cliques of an edge-clique cover that contain v , restricted to $N(v)$, cover all vertices of $N(v)$ by cliques. Assign each neighbor to one such clique containing it and discard empty parts. This gives a clique partition of $N(v)$ with at most as many parts as cliques through v . Minimize over edge-clique covers. $\square$

An upper certificate consists of a proper coloring and an actual edge-clique cover. Each clique is checked pairwise, every edge must be covered, and vertex incidences are counted individually. For 1 219 of the 1 252 cones, including all 171 equality graphs, Lemma 3 matches the load of the supplied cover and hence certifies the exact local cover number. The remaining 33 cones have $\text{lcc} = 3$, whereas all their neighborhood partition bounds are at most two.

2.2 The remaining lower certificates

For these 33 cases we use finite directed acyclic graphs certifying that no cover of load at most two exists. For a fixed candidate load k , a search state consists of uncovered edges U and current vertex loads (ℓ_v) . The root has $U = E(G)$ and all loads zero. At a state with $U \neq \emptyset$, choose an edge $e \in U$. For every clique Q containing e with $\ell_v < k$ for all $v \in Q$, the state has a child

$$U' = U \setminus E(G[Q]), \quad \ell'_v = \ell_v + \mathbf{1}_{v \in Q}. \quad (4)$$

All cliques of size at least two are considered, including nonmaximal ones.

Lemma 4. *Suppose a finite collection of states contains the root, every state has nonempty U , and the children of each state are exactly all legal transitions in (4). Then G has no edge-clique cover of load at most k .*

Proof. Every transition strictly decreases $|U|$, since e is removed. If a feasible completion existed at a state, some clique of that completion would cover e . Adding it respects the load bound and therefore gives one of the required children with a feasible completion. A previously selected clique cannot contain an edge still uncovered, so omitting the selected-clique history from the state does not omit any necessary branch. Following a feasible completion would eventually reach $U = \emptyset$, contradicting the hypotheses. Leaves with no legal choice are therefore infeasible, as is the root. $\square$

The saved certificates comprise 33 such graphs with 2 258 states in total at $k = 2$. A verifier independently enumerates the full clique set and checks exact equality of the legal child-state set with the saved set at every node. Together with explicit 3-local covers, this proves all 33 remaining optima. Auxiliary SAT search was used to obtain some data, but no solver status, floating-point result, or SAT proof file is used in these lower-bound checks.

3 Exhaustion of the graph domain and proof of the theorem

Let D_n be the supplied representative set of graphs of order n , starting with the unique one-vertex graph. From every graph in D_{n-1} , add a vertex with each of its 2^{n-1} possible neighborhoods. Every n -vertex graph is isomorphic to one of these extensions: delete a vertex and apply the inductive completeness of D_{n-1} , carrying its neighborhood through the isomorphism. This argument includes disconnected and edgeless graphs.

The domain verifier matches every extension to a member of D_n by an explicit adjacency-preserving bijection, then directly checks adjacency and nonadjacency for every pair. The program `labelg` proposes the target representative; its proposed identification is always checked. Distinct representatives are proved nonisomorphic as follows. First group them by the invariant consisting of the sorted vertex signatures

$$(\deg(v), \text{multiset of degrees of neighbors of } v).$$

Within every group, test every pair by exhaustive bijection search. The search tries each unused image vertex of the appropriate signature and tests adjacency to every already mapped vertex. These are necessary conditions for an isomorphism, so the search is exhaustive. Different groups cannot be isomorphic. Across the domain this required 4 000 pair tests; at order eight all 133 632 vertex extensions were matched. Every supplied representative is reached. Thus the domain has exactly the class counts in Table 1, without relying on the counts alone as evidence of completeness.

The same verifier gives checked bijections from the 1 252 cone rows onto exactly those domain representatives having a universal vertex. It also verifies that the 12 345 noncone rows give exactly the complementary subset. For each noncone row a proper coloring with a colors and an edge-clique cover of maximum load b satisfy

$$\chi(G) + \text{lcc}(G) \leq a + b \leq n. \quad (5)$$

Optimality of these two witnesses is unnecessary. In particular, edgeless graphs are excluded by an empty cover and a one-color assignment.

Proof of Theorem 1. The preceding induction and bijection checks cover every graph in the stated range exactly up to isomorphism. For noncones, (5) excludes equality. For each cone, both subset tables in Lemma 2 are checked on every subset; a proper coloring realizes $c(V(G))$. An explicit cover gives the upper bound for $\text{lcc}(G)$, and Lemma 3 or Lemma 4 gives the matching lower bound.

In every cone row, the last vertex is checked to be universal, and its deletion is checked edge by edge against the recorded residual H . Writing $S = V(G) \setminus \{u\}$, the verifier compares the two Boolean conditions

$$[\text{lcc}(G) + c(V(G)) = n + 1] \quad \text{and} \quad [c(S) + p(S) = n].$$

They agree on all 1 252 cone rows. Since $p(S) = \chi(\overline{H})$ and $|V(H)| = n - 1$, this is exactly (1) on cones. The row-to-domain bijections transfer the decisions to all graphs of the specified orders. Extracting the true rows yields precisely the 171 listed representatives and the counts in Table 1. $\square$

For example, the first three representatives are K_2 , P_3 , and K_3 . Their pairs (χ, lcc) are $(2, 1)$, $(2, 2)$, and $(3, 1)$, respectively. The two edges of P_3 must appear separately and give load two at its middle vertex; these are also the two cliques required by its independent neighborhood.

4 Reproduction and limitations

The accompanying archive `supplement.zip` contains the decisive data and two certificate verifiers. After extracting it into a fresh directory, run

```
python evidence/verify_certificates.py
python evidence/verify_domain.py
```

Use Python 3.10 or later with assertions enabled; the recorded successful environment used Python 3.13.5. The first checker uses only the Python standard library. The second additionally needs `labelg` on the executable search path; its graph identifications are verified by the bijection checks described above. Neither command needs a SAT solver or NetworkX. Both commands regenerate some derived JSON files, which is why they should be run on an extracted copy. The search environment also used NetworkX 3.4.2, python-sat 1.9.dev15, and the graph tools `geng` and `labelg`; replay of the decisive certificates does not repeat that search.

The main data files, under the archive’s `evidence/` directory, are `cone-results.jsonl`, `noncone-exclusions.jsonl`, `lcc-lower-dags.json`, `equality-representatives.json`, and `domain-1.g6` through `domain-8.g6`. The equality file records `graph6`, `residual_graph6`, `chi`, `lcc`, a coloring, `cover_masks`, and a neighborhood lower certificate. A mask q represents the vertex subset $\{v : \text{bit } v \text{ of } q \text{ is } 1\}$, with vertices numbered from zero. Complete subset tables for the same representatives occur in the cone file. The lower-certificate vertex need not be the appended universal vertex.

A separate AI-assisted mathematical review replayed both verifiers on copies and used a separately written direct color-assignment search to recompute every cone chromatic number, the required neighborhood partition bounds, and the two residual chromatic quantities. It also recomputed the 33 load-two infeasibilities without reading their saved state graphs, using a different uncovered-edge choice and exploring 10015 states in total. This is a separate software and review pass, not a claim of statistical independence, external peer review, or proof-assistant formalization.

Exact enumeration with independently checkable graph6 certificates also appears in Zeng’s work on the eight-vertex whiskered-graph problem [2]. That reference provides adjacent computational methodology; none of its algebraic conclusions is used to establish the local cover classification here.

All decisive operations are finite Boolean or integer calculations. There is no sampling, numerical tolerance, or extrapolation. The conclusion remains a computer-assisted theorem with ordinary software execution as part of its verification. The original finite task is fully covered, while the all-order problem remains outside this result. The motivating preprint was inspected locally, but attempts to refresh online sources and search for equivalent classifications on 9 September 2026 failed at name resolution. Accordingly, priority and the absence of an equivalent finite classification have not been established; no such claim is made.

Computational and AI assistance. OpenAI Codex assisted with problem analysis, exact enumeration and certificate code, a separate mathematical review, and manuscript preparation. The mathematical basis of the finite result is the elementary arguments above together with the supplied exhaustive data and their explicit verification, rather than an AI assessment or a successful document build.

References

- [1] Akbar Davoodi, *Local clique covers and chromatic number*, arXiv:2609.07988v1, 7 September 2026. <https://arxiv.org/abs/2609.07988v1>.
- [2] Zijian Zeng, *The Whiskered-Graph Weak Lefschetz Conjecture for Graphs on Eight Vertices*, 2026. <https://doi.org/10.2139/ssrn.7385138>.

A Complete equality representatives

Table 2 gives the complete list, grouped by n and χ ; within a row, every graph has the displayed $\text{lcc} = n + 1 - \chi$. The strings are graph6 encodings of G , rather than of its residual graph. For these orders, decoding is elementary: the first character has ASCII value $63 + n$; subtract 63 from each remaining character and read its six binary digits from most to least significant. These bits specify edges in the order $(0, 1), (0, 2), (1, 2), (0, 3), (1, 3), (2, 3), \dots$, with zero padding after the final pair. The last vertex in each listed encoding is universal. These representatives need not be canonical graph6 strings under another labeling convention.

Table 2: All 171 equality representatives.

n	χ	lcc	graph6 representatives
2	2	1	A_
3	2	2	BW
3	3	1	Bw
4	2	3	CF
4	3	2	CN, Cv
4	4	1	C~
5	2	4	D?{
5	3	3	D@{, DB{, DF{
5	4	2	DJ{, DN{, D {
5	5	1	D~{
6	2	5	E?Bw
6	3	4	E?Fw, E?~w, EBFw, EgBw
6	4	3	E@~w, EF~w, EJFw, Ehfw, EjZw, EwBw, Exbw, E Bw
6	5	2	EJ~w, Eevw, Ef~w, En~w
6	6	1	E~~w
7	2	6	F??Fw
7	3	5	F??Nw, F?Bvw, F?B~w, FCaFw, FI?Fw
7	4	4	F?F~w, F?~~w, F@c^w, FBC^w, FB{Fw, FB{Nw, FCaNw, FCa^w, FC{Vw, FJ?Fw, FhXfw, FhcFw, FzWfW, FzWfw
7	5	3	FCe^w, F_~~w, Fhf~w, Fjt^w, Fn{Fw, Fn{Nw, F~@fw, F~AFw, F~Hfw, F~Xvw, F~_Vw, F~`fw, F~aFw, F~zfw
7	6	2	Fz~~w, F~{Fw, F~{^w, F~ Fw, F~~Nw
7	7	1	F~~~~w
8	2	7	G???F{
8	3	6	G???N{, G??F~{, G?BwF{, GH??F{, G_ACF{, GgP?F{
8	4	5	G?Bc~{, G?B~~{, G?~wF{, G?~wN{, G@G`N{, GG?`N{, GHG`N{, GI+AF{, GJO`F{, GOg?F{, GOgKF{, GTg`F{, GWJGF{, Geo?F{, Ghc?F{, GixGF{, GjW?F{, GjW@F{, GmpAF{, GsaCf{, GupAF{, GzW`F{, GzWaF{
8	5	4	GD^W~{, GFC^~{, GF~w~{, GG@bN{, GHXgN{, G_~wF{, G_~wV{, G_~yF{, GetAF{, GgB~~{, GgxfF{, GhfwF{, GhfwN{, Gh JV{, Gj[?F{, GjtWF{, GjtWV{, GjtYF{, Gn{?F{, Gn{GF{, Gn{GV{, Gn{OV{, G}BFn{, G}~IF{, G}~KV{, G~H_F{, G~H`F{, G~HaF{, G~XoF{, G~Xof{, G~XqF{, G~_QF{, G~`_F{, G~`aF{, G~`cF{, G~a@F{, G~aCF{, G~zDF{
8	6	3	G@z~{, G@ ~{, GB ~{, GEn~~{, GEv~~{, GF~{~{, Gsn]~{, Gz~wF{, Gz~yF{, G}vUn{, G}vnf{, G~znV{, G~{?F{, G~{OV{, G~{WF{, G~{WV{, G~{Wv{, G~{sV{, G~ ?F{, G~ AF{, G~~BF{, G~~IF{, G~~vf{
8	7	2	G~~~~{, G~~]~{, G~~wF{, G~~xF{, G~~zF{, G~~}N{
8	8	1	G~~~~{